

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: GEN-301 — Digital Logic and Processors

Time: Three hours

Maximum Marks: 70

Instructions: Answer all Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A: Long Answer Questions

[15 × 2 = 30]

Instructions: Long answer type questions to be answered in about 750 words each. Attempt any two.

Question 1. Prove the following Boolean identities:

- (a) $AC + ABC = AC$
- (b) $A + \bar{A}B = A + B$
- (c) $AB\bar{C} + A\bar{B}C + ABC = A(B + C)$

OR**Question 2.** Explain about signal processing and discuss its different types in detail.**Question 3.** Convert the number system from one form to another:

- (a) $(6)_{10} = (\quad)_2$
- (b) $(110010)_2 = (\quad)_8$
- (c) $(A9)_{16} = (\quad)_{10}$

OR**Question 4.** What are the basic gates in a logic circuit? Give a brief explanation of each with its respective truth table.**SECTION — B: Short Answer Questions**

[5 × 6 = 30]

Instructions: Attempt any six questions. Short answer type questions to be answered in about 550 words if theoretical.

Question 1. What is a combinational circuit? Design a combinational circuit with two AND gates and one OR gate, and draw its truth table.**Question 2.** Perform the following binary mathematical operations:

(a) Binary Addition:

$$\begin{array}{r} 1101_2 \\ + 1010_2 \\ \hline \end{array}$$

(b) Binary Subtraction:

$$\begin{array}{r} 1111_2 \\ - 101_2 \\ \hline \end{array}$$

Question 3. What will be the output for the given logic circuit if $X_1 = 1$ and $X_2 = 0$?

Question 4. Calculate the number of cells in a Karnaugh Map (K-map) if we have 3 variables. Draw the K-map structure and write the cell representation.

Question 5. Simplify/Demorganize the following expression:

$$\overline{(A + B)(C + D)}$$

Question 6. Convert the given logical expression into Sum of Products (SOP) form:

$$ABC + A\bar{C}D + ABCD$$

Question 7. What are latches and what are their types? Explain the NAND latch in detail.

Question 8. What is a multiplexer? Explain its working principles.

SECTION — C: Very Short Answer Questions

[10 × 1 = 10]

Instructions: Answer the following questions in a single word or sentence.

1. In K-map, what does the letter 'K' stand for? (Karnaugh)
2. Give one example of a Universal Gate.
3. What type of arithmetic operation is performed by an OR gate?
4. "A binary variable can store the value of zero or one." Is this statement true or false?
5. Which device converts a single input into multiple outputs? (Demultiplexer)
6. Does a flip-flop store 1-bit of information?
7. Is a register used for storing data?
8. Multiply $(1001)_2$ with $(10)_2$.
9. Write the full form of MSB. (Most Significant Bit)
10. Write the 1's complement of $(1000)_2$.

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